VESC Setup Manual

RC Car Assembly Manual Series No.2

This manual is the second one of our RC car assembly manual series. In this manual, we will explain the VESC setup process with VESC tool. Please read and follow the instructions carefully. There may be changes in parts due to future discontinuation issues.

RC Car Assembly Manual Series

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1. RC Car Hardware Assembly Manual
2. VESC Setup Manual
3. Jetson Xavier NX Setup Manual
4. Keyboard Control with ROS2
5. RC Car Calibration Manual

RLLAB

http://rllab.snu.ac.kr

Last modified: 24.08.12

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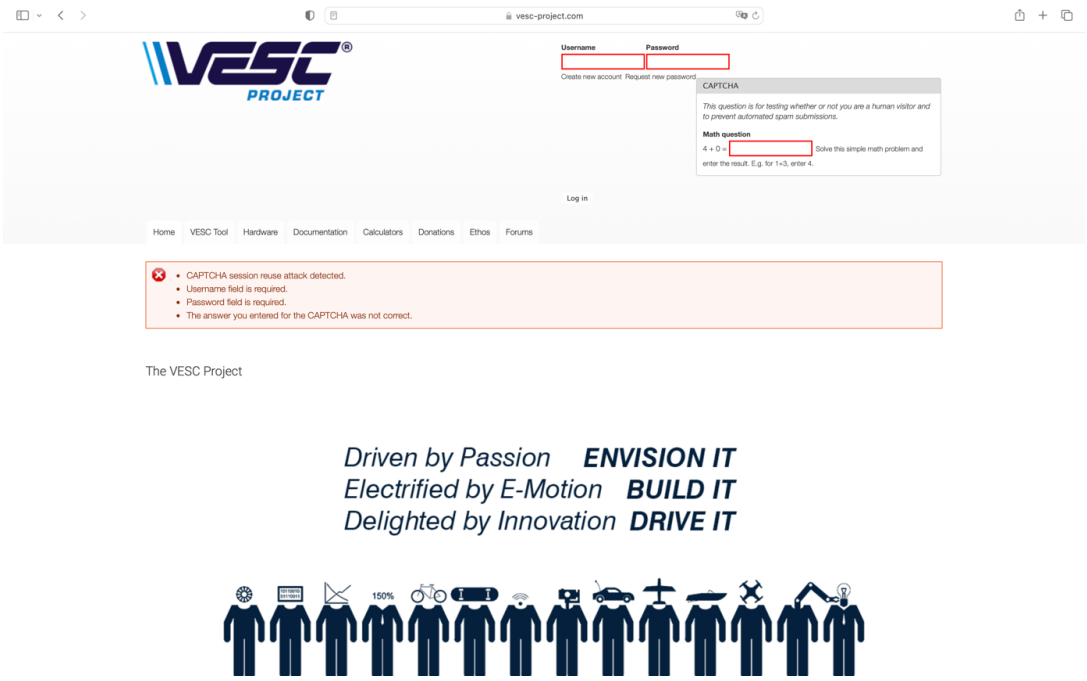
0. Prepare Linux laptop
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0. Prepare Linux laptop

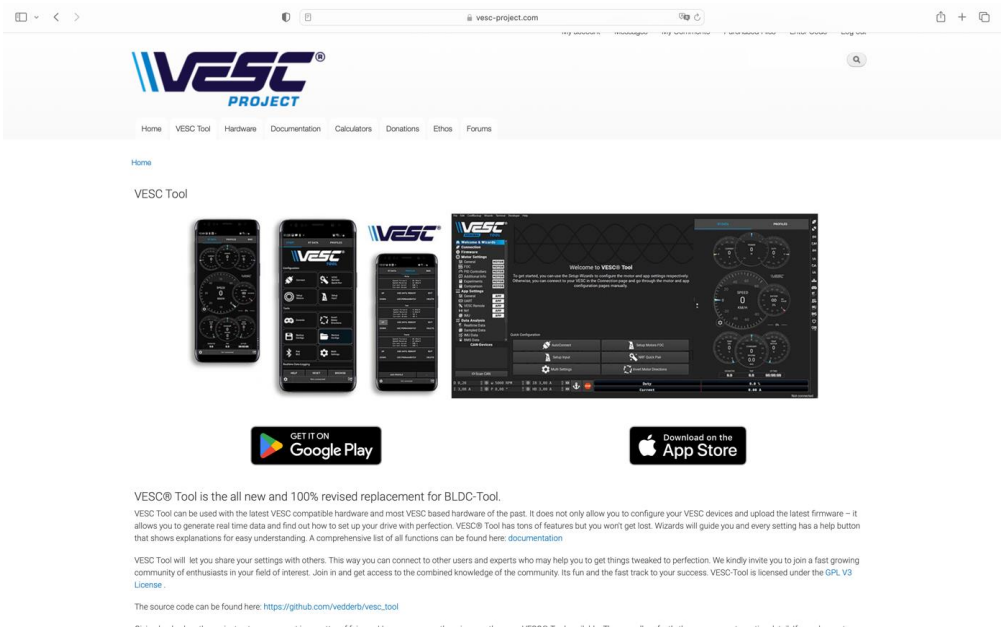
You have to prepare linux laptop before the class.

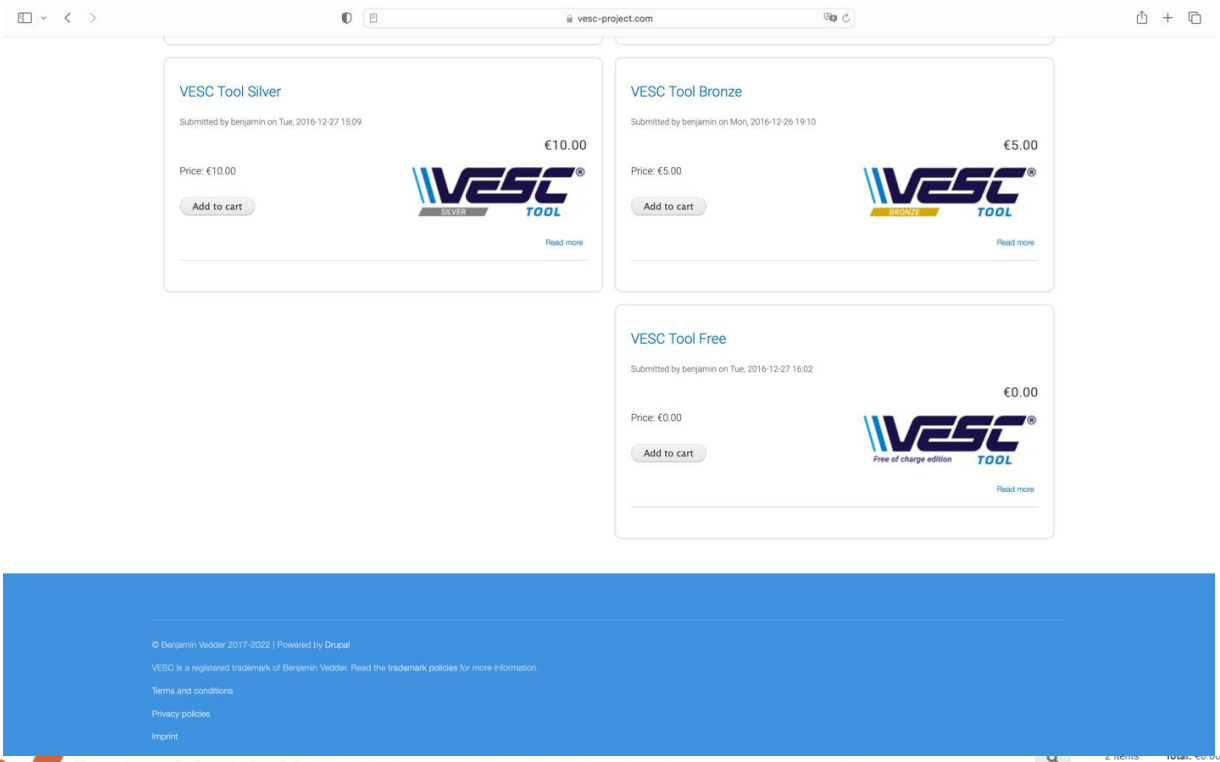
1. Download VESC tool

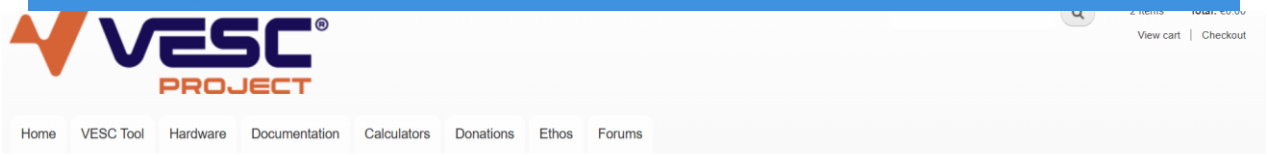
Create an account on the vesc-project website (https://www.vesc-project.com/)



Navigate to VESC tool → Add to cart “VESC Tool Free” → Checkout → Review order → My account → Files → Download zip file for your OS (Linux is strongly recommended)







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File downloads

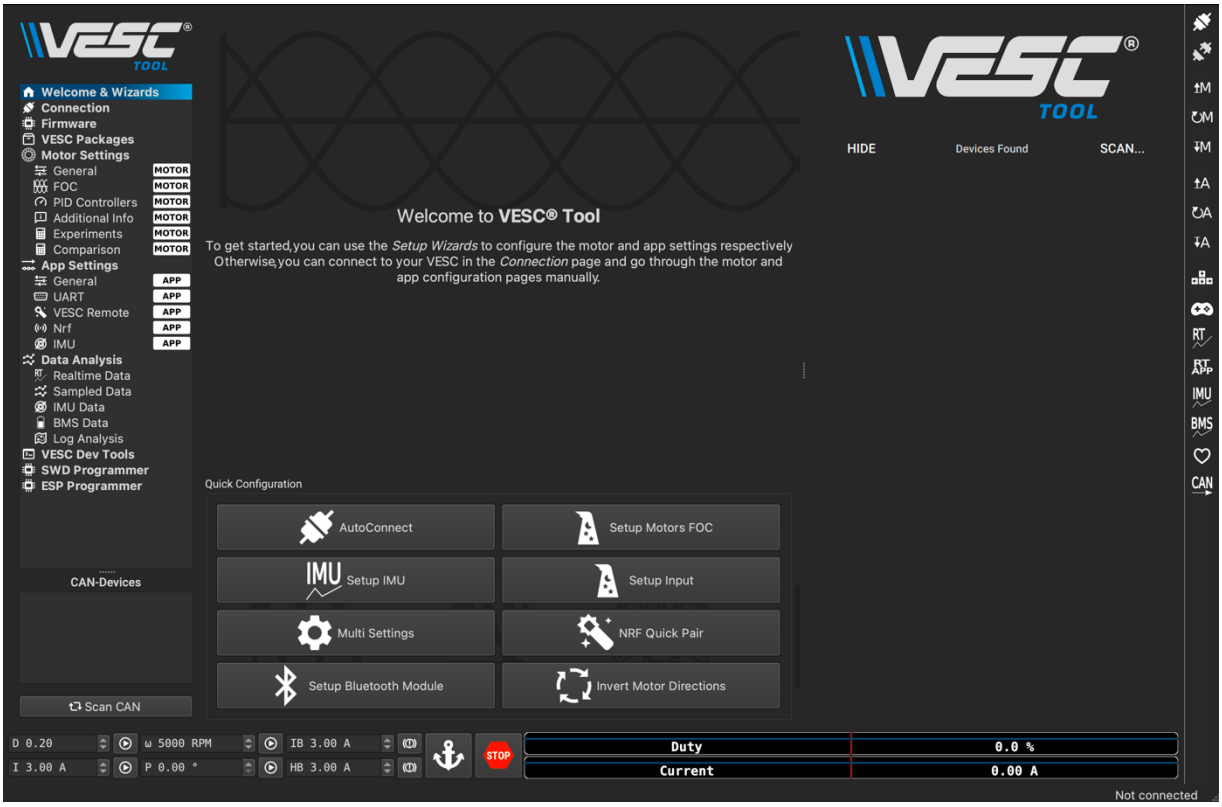
View Edit Files Orders File browser

Purchased	Filename	Description	Downloads	Addresses
2026-09-30	vesc_tool_free_linux.zip (expires on 2027-03-30)		0/Unlimited	0/Unlimited
2026-09-30	vesc_tool_free_windows.zip (expires on 2027-03-30)		0/Unlimited	0/Unlimited
2026-09-30	vesc_tool_android.zip (expires on 2027-03-30)		0/Unlimited	0/Unlimited
2026-09-30	vesc_tool_mac.zip (expires on 2027-03-30)		0/Unlimited	0/Unlimited
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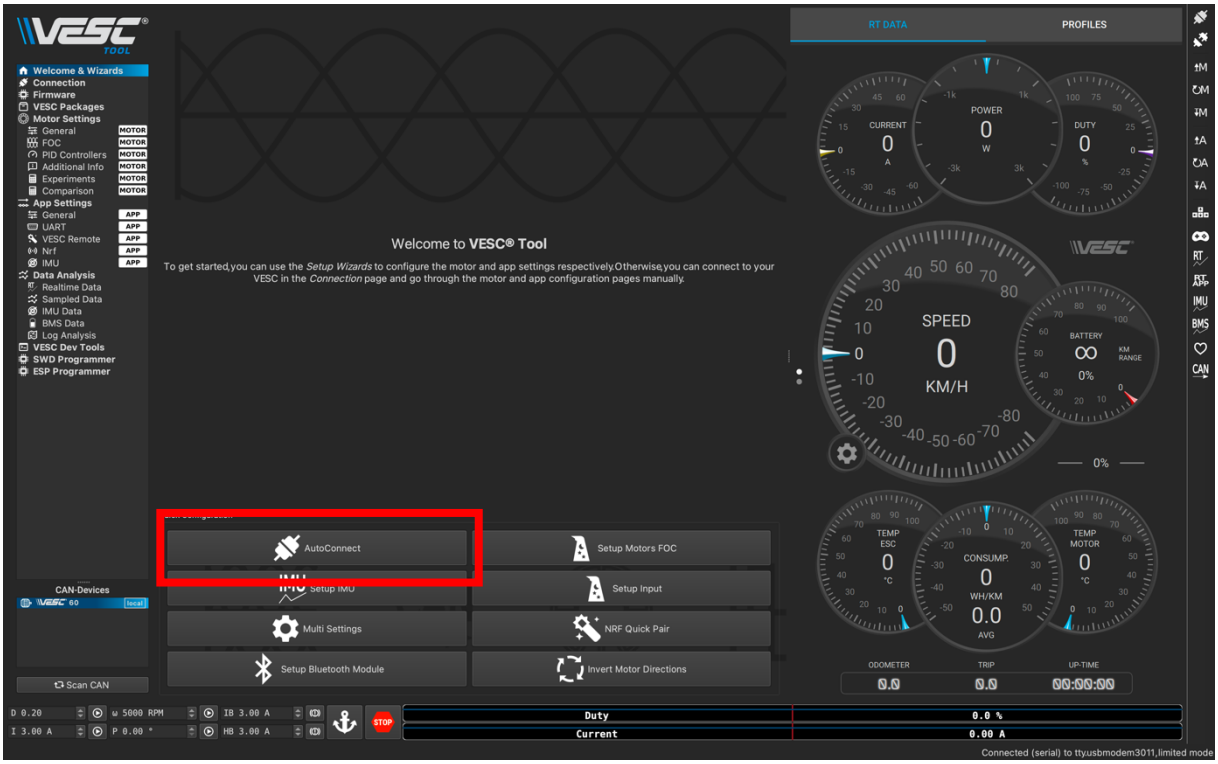
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Do not use any download acceleration feature to download the file, or you may lock yourself out of the download.

2. Firmware update

Turn on the VESC tool

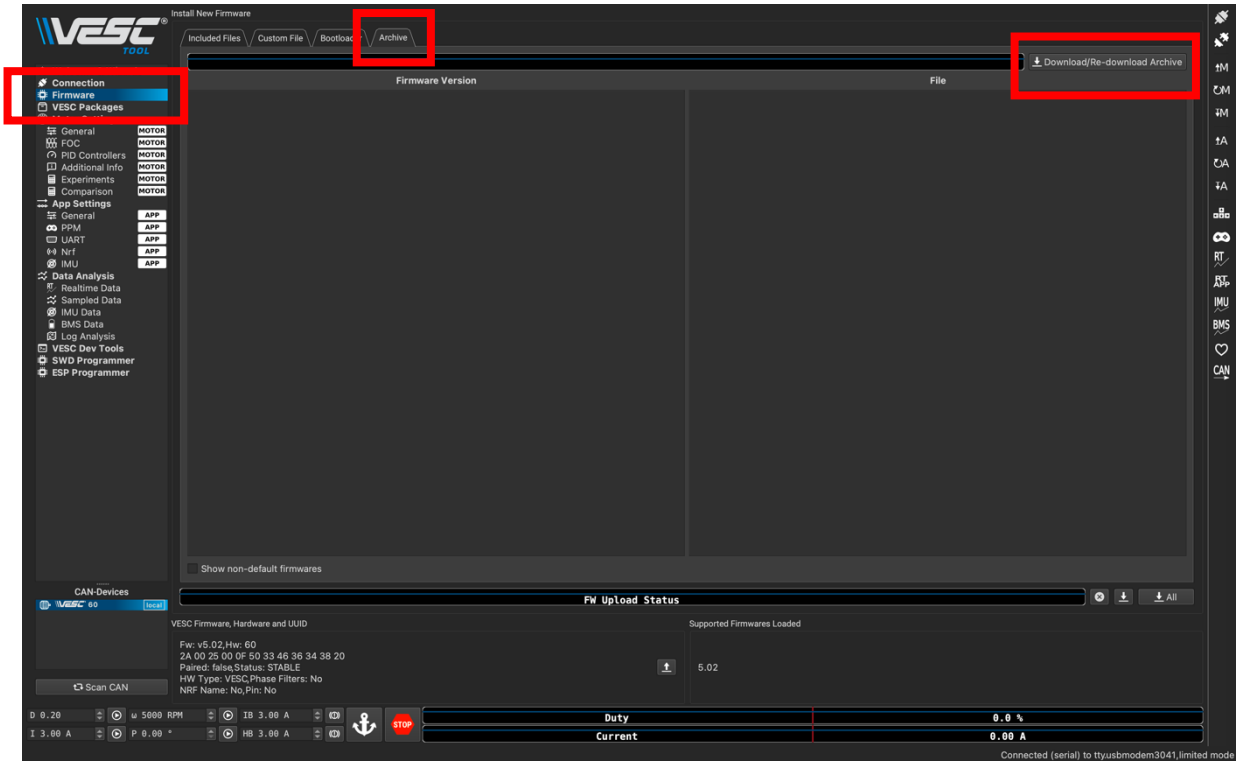


Click AutoConnect. If properly connected, you will see a screen resembling a dashboard.

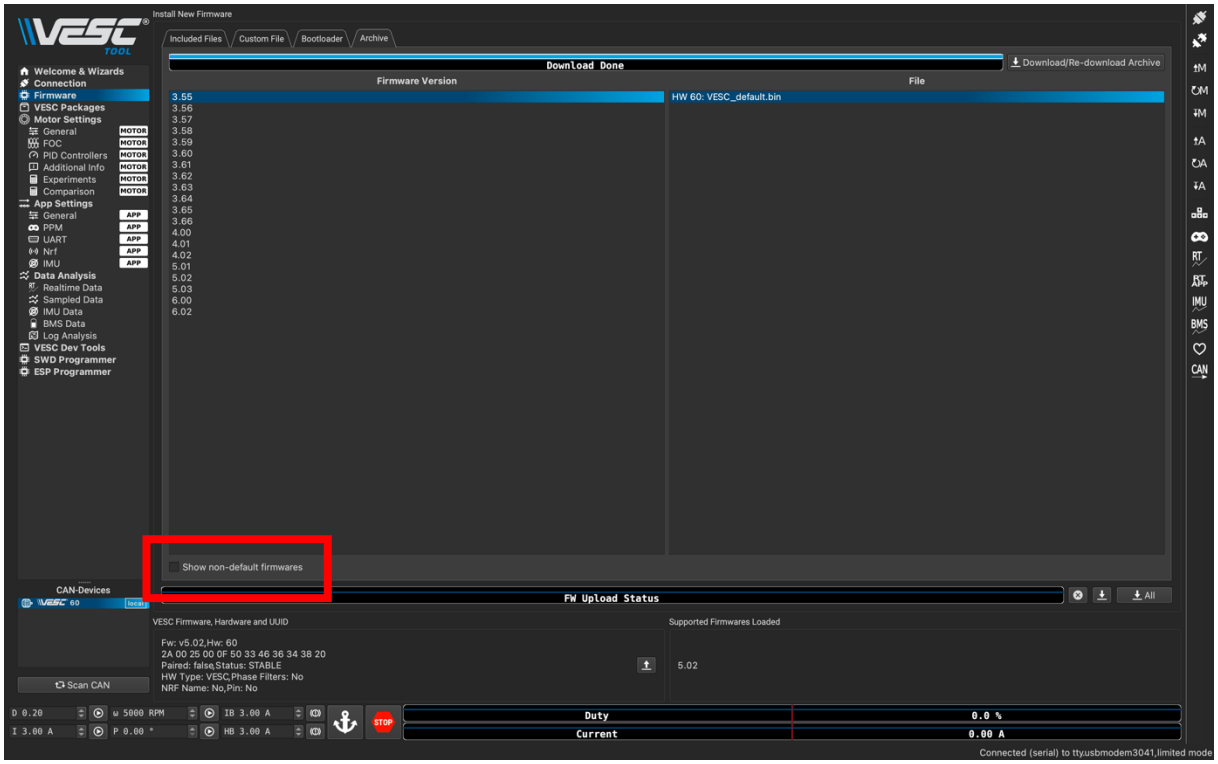


Change the firmware. This is to find firmware that allows the VESC's PWM terminal to send control signals for the servo motor.

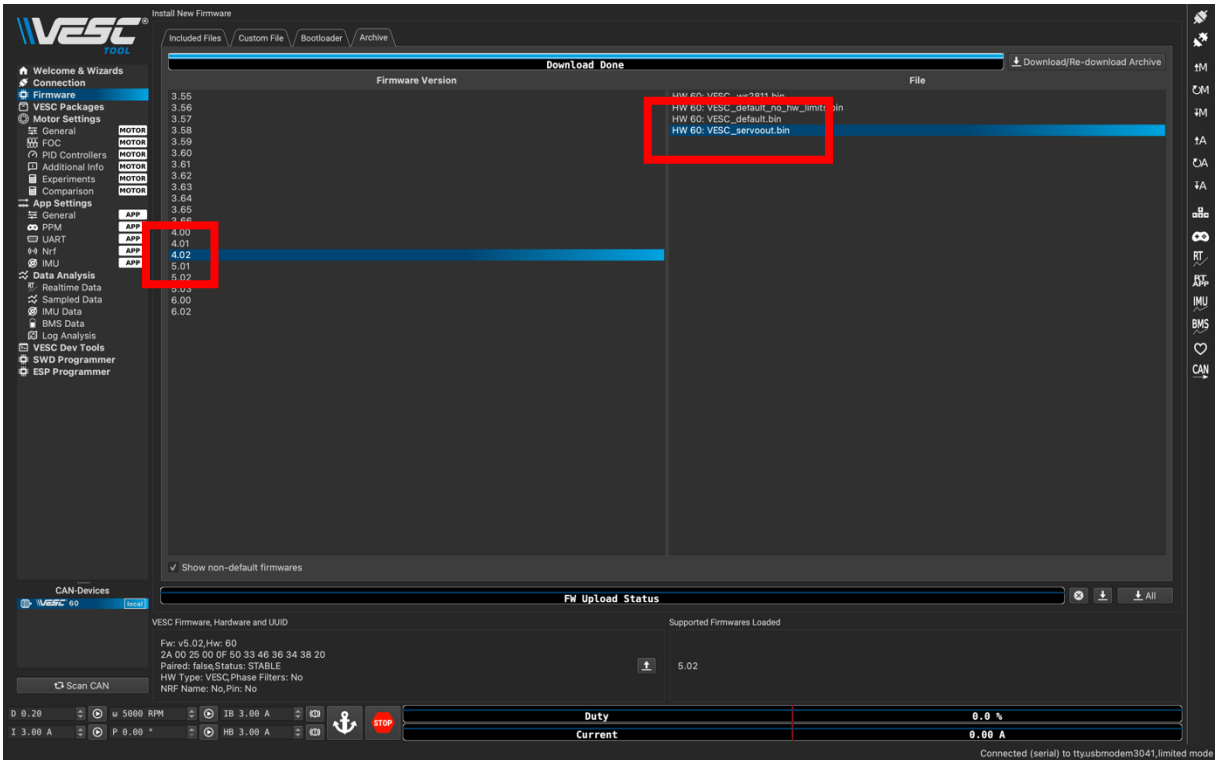
Navigate to “firmware” in the left side. Select “Archive” tab and click “Download/Re-Download Archive.”



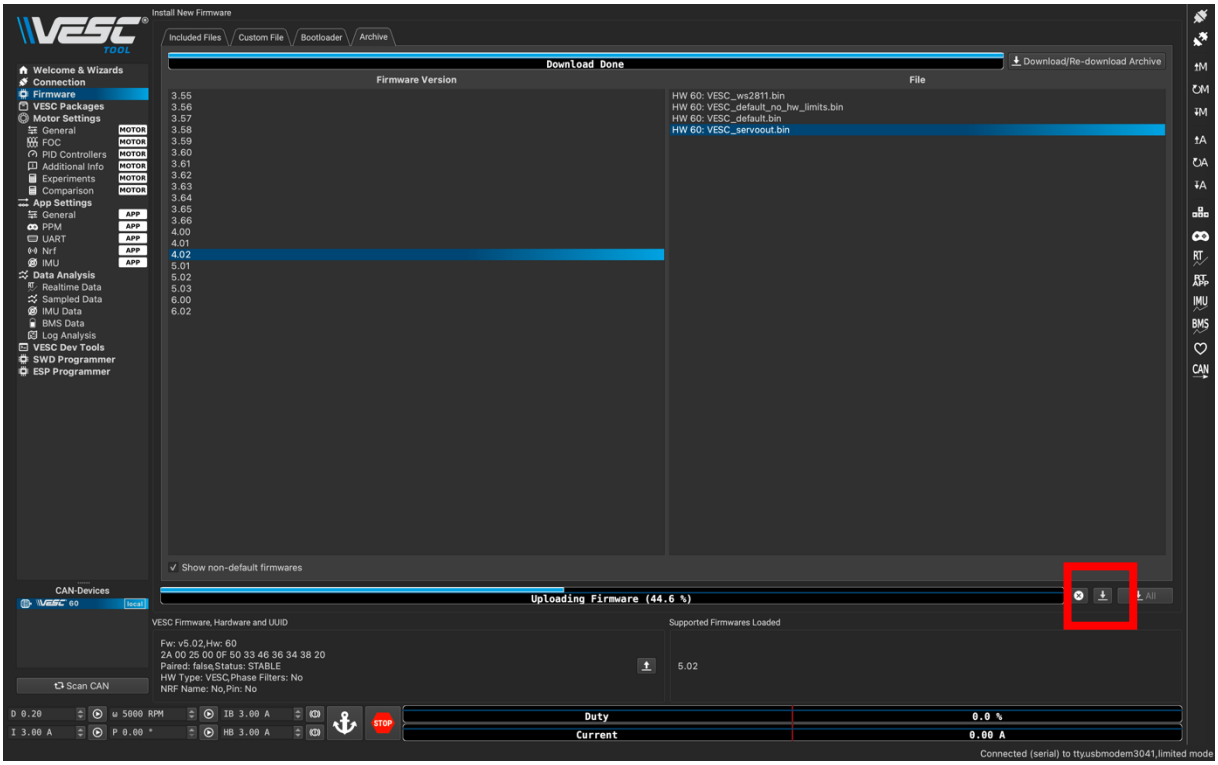
Click “Show non-default firmwares”



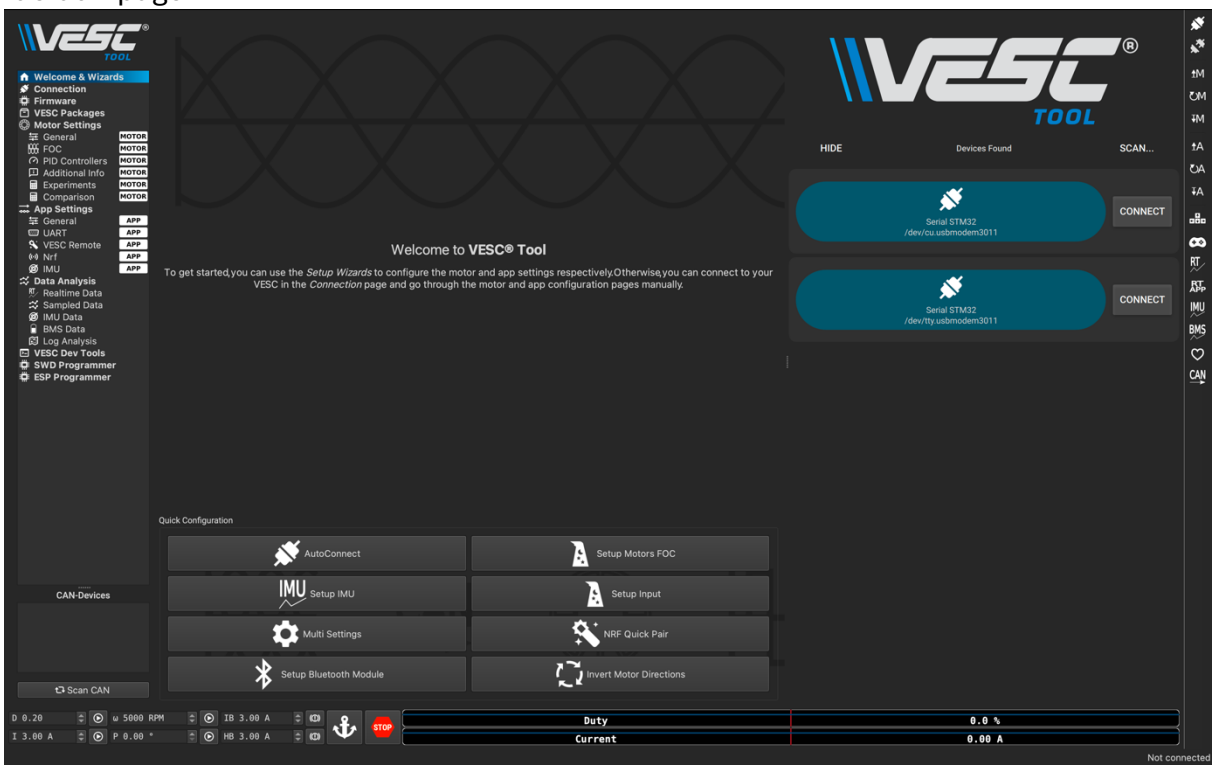
Select “VESC_servoout.bin” included in version 4.02.



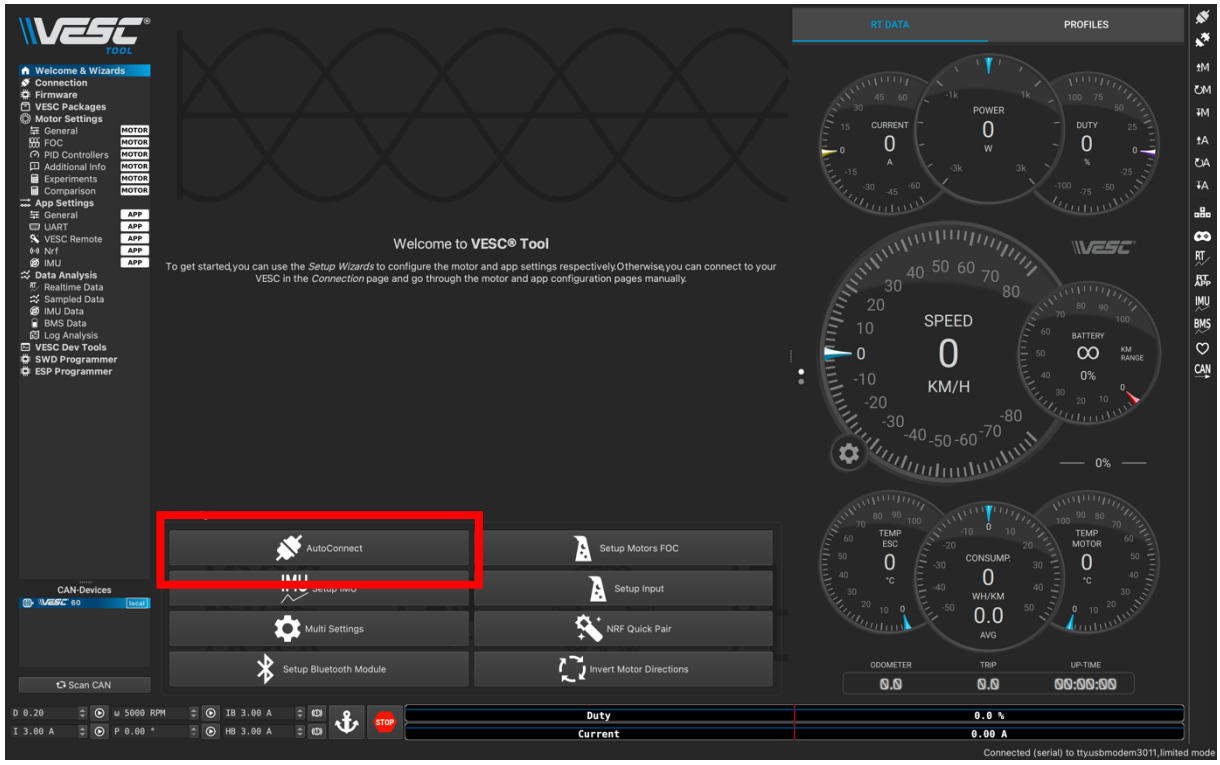
Upload firmware by clicking the button below.



Come to default page



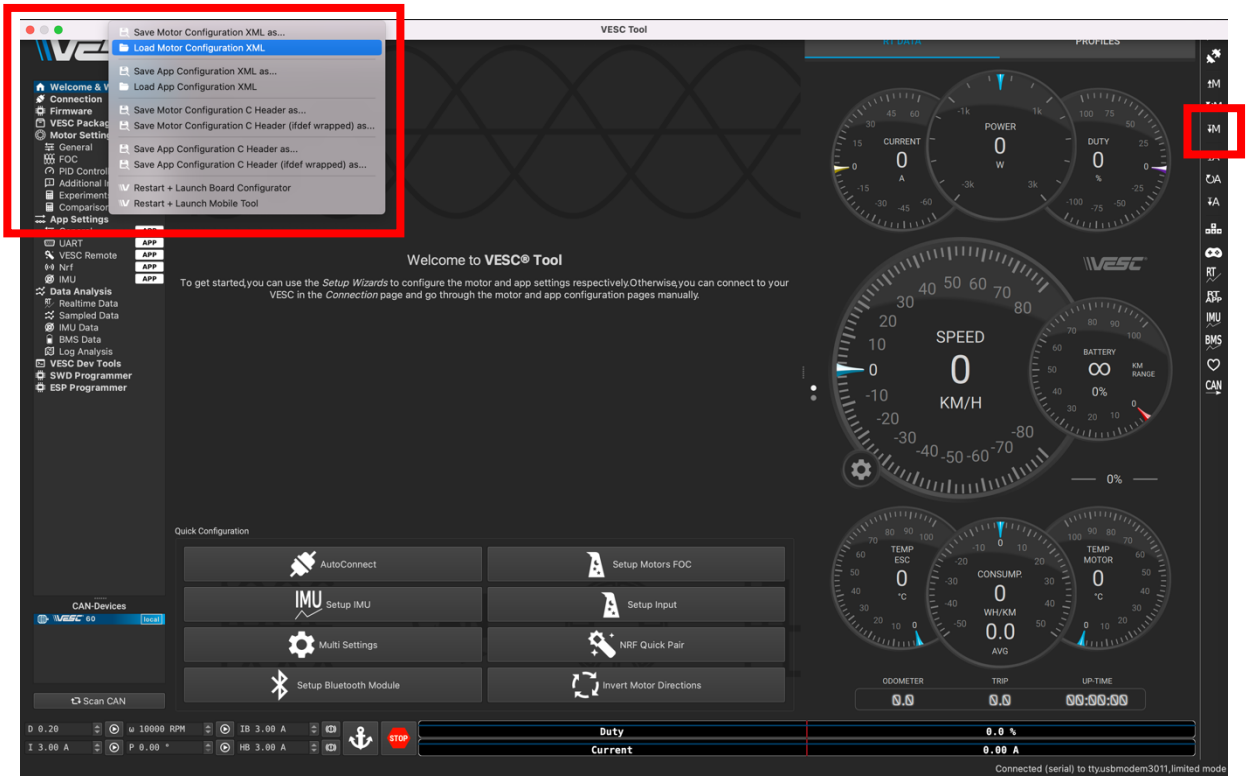
Click “AutoConnect”



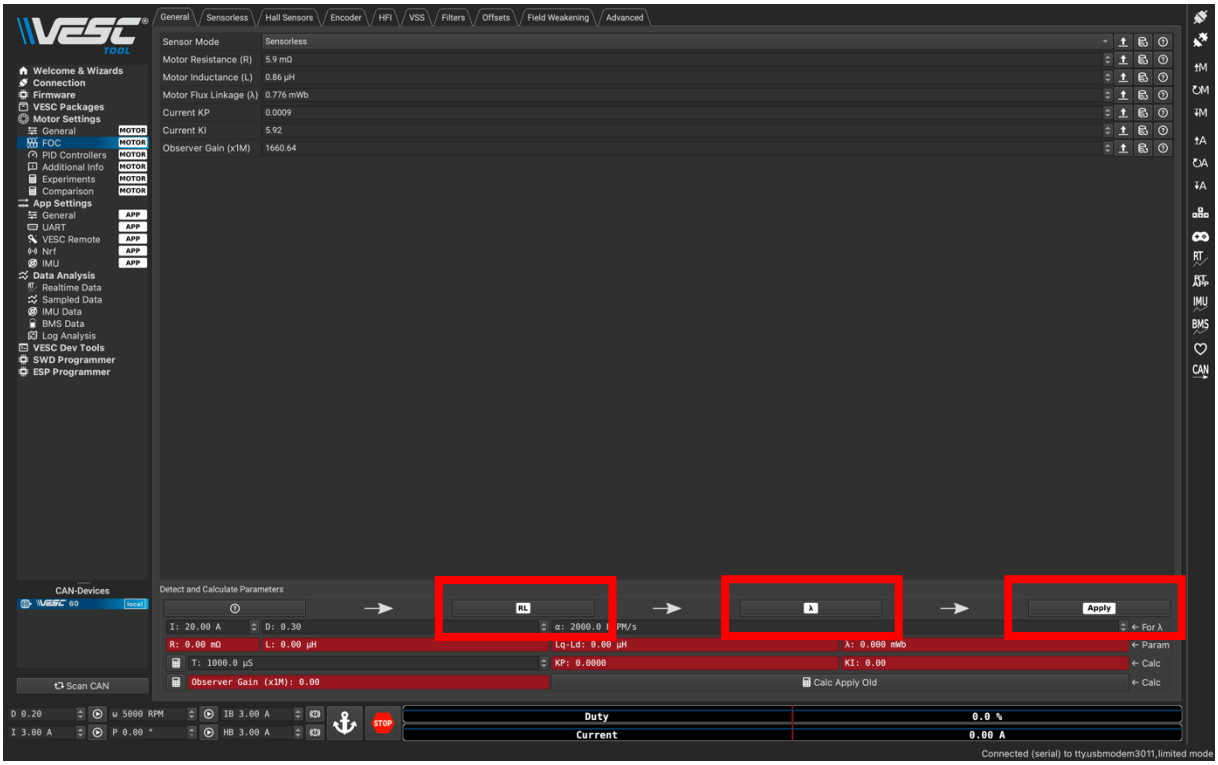
3. Setting Up Motor Configuration

First, Download xml file from the link (https://drive.google.com/file/d/1-KiAh3hCROPZAPeOJtXWvfxKY35lhhTO/view).
Upload xml file by selecting “file” → “Load motor configuration XML”.

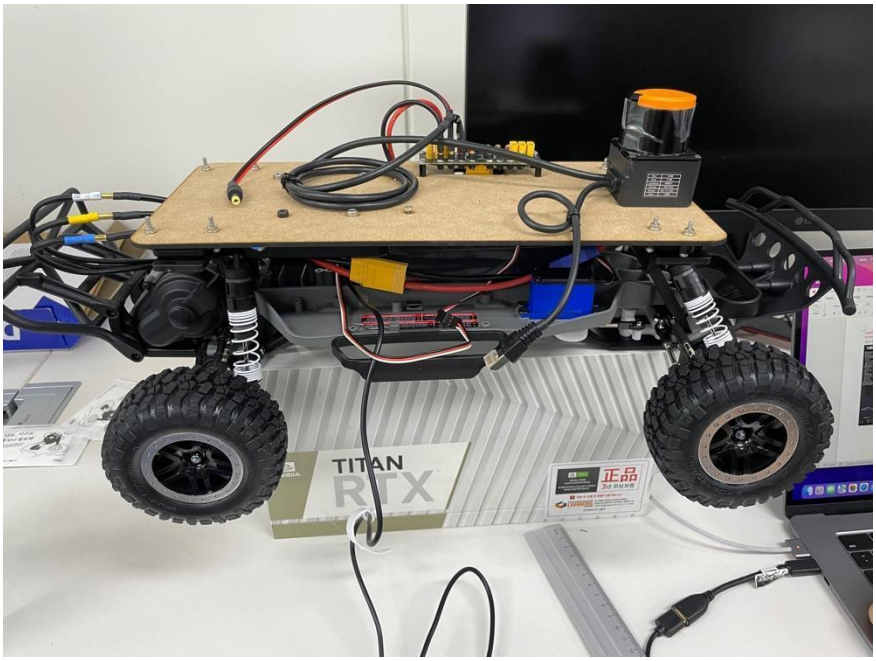
After the XML is uploaded, click on the **Write Motor Configuration** button (the button with a down arrow and the letter M) on the right side of the screen to apply the motor configuration. Note that in the future, you’ll have to press this button whenever you make a change in motor configuration.



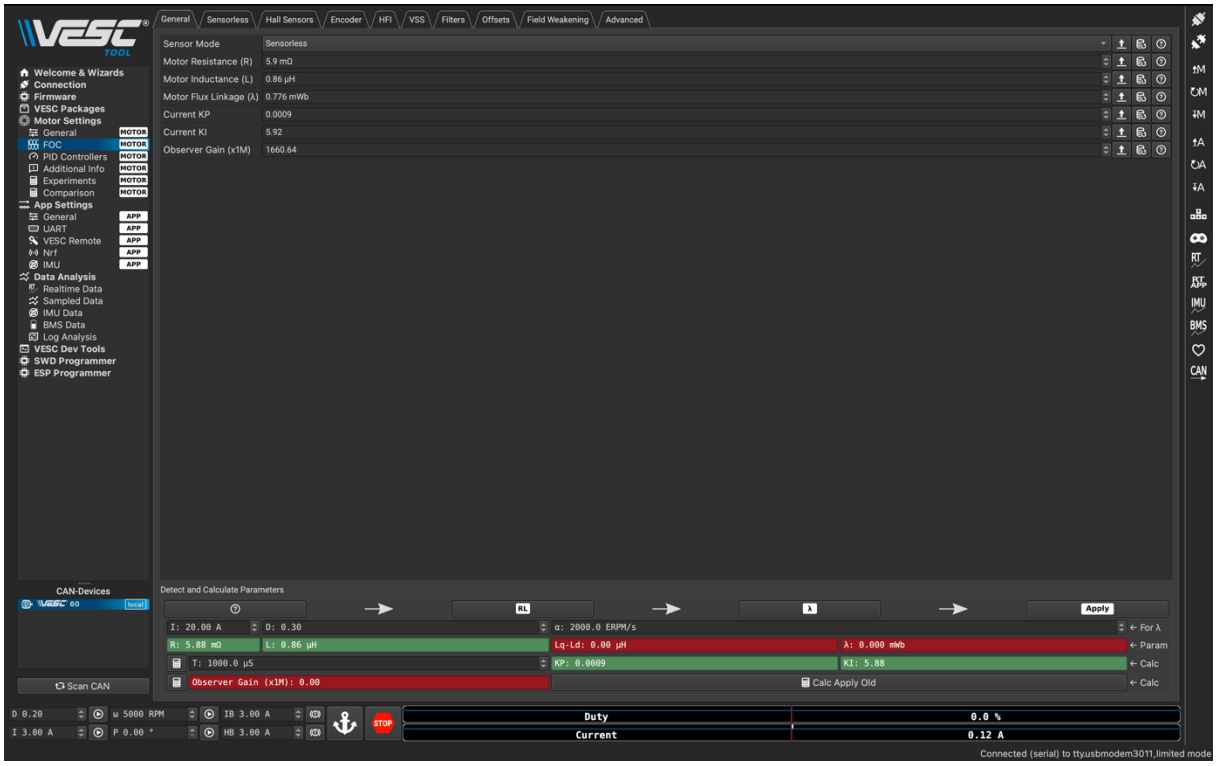
Navigate to “Motor Settings” → “FOC” in the left side.
Initial setup is as below.



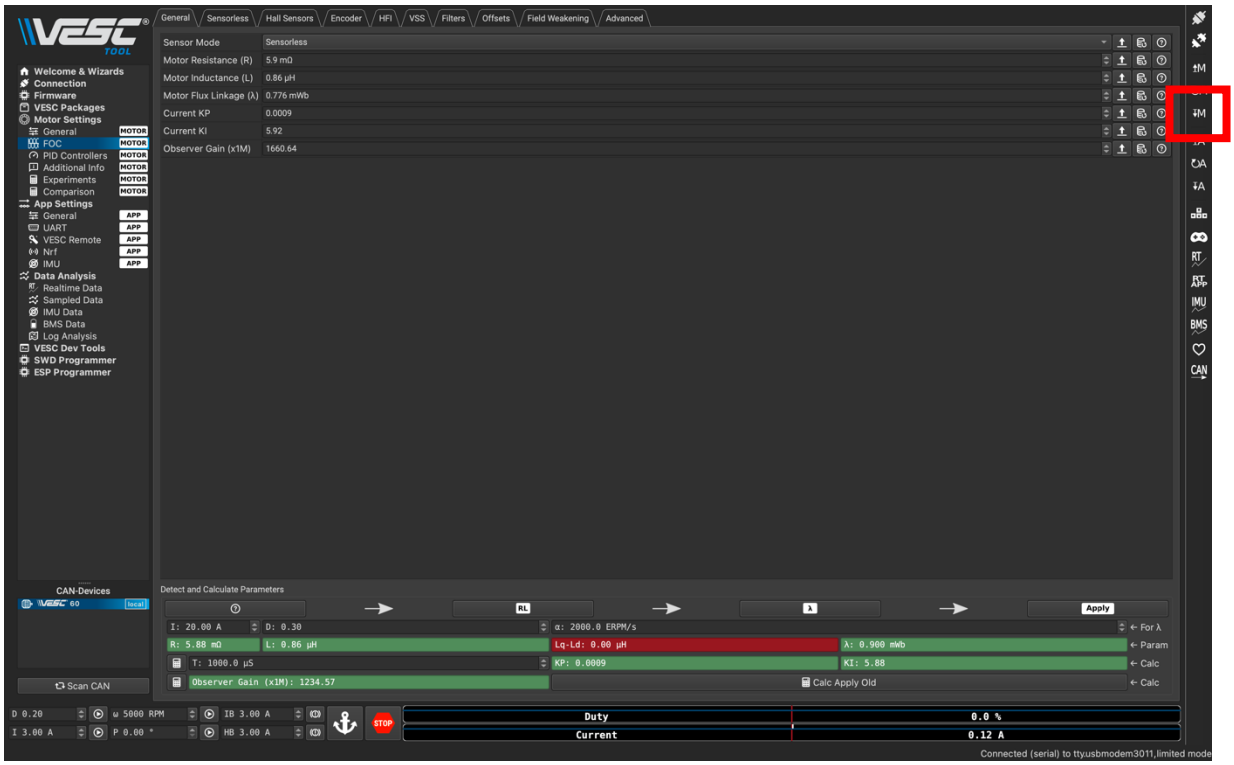
Click “RL” → “\lambda” → “Apply” in order.
Caution: Note that during the measuring process, the motor will make noise and spin, make sure the wheels of your vehicle are clear.
Example)



After clicking “RL”

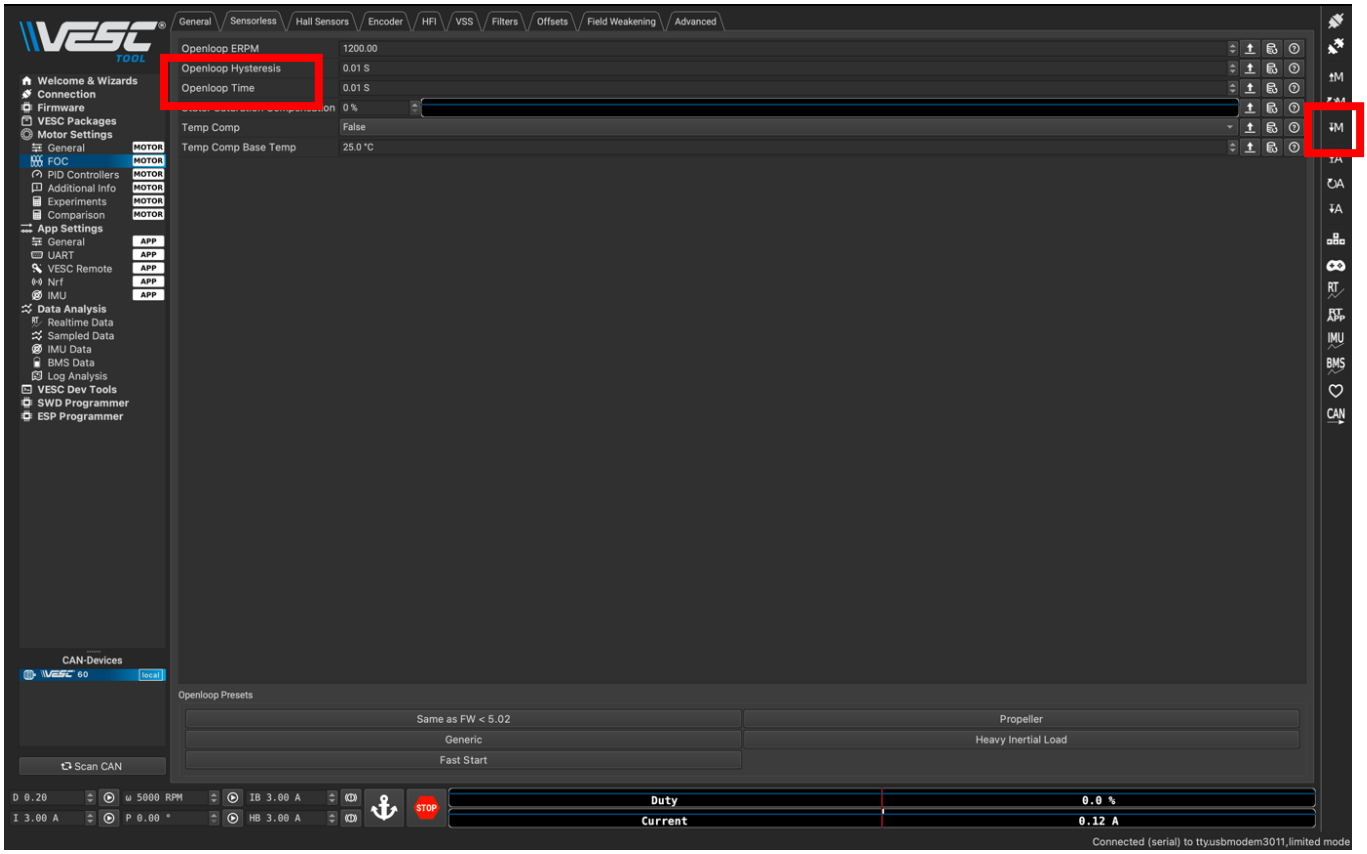


After clicking “\lambda”

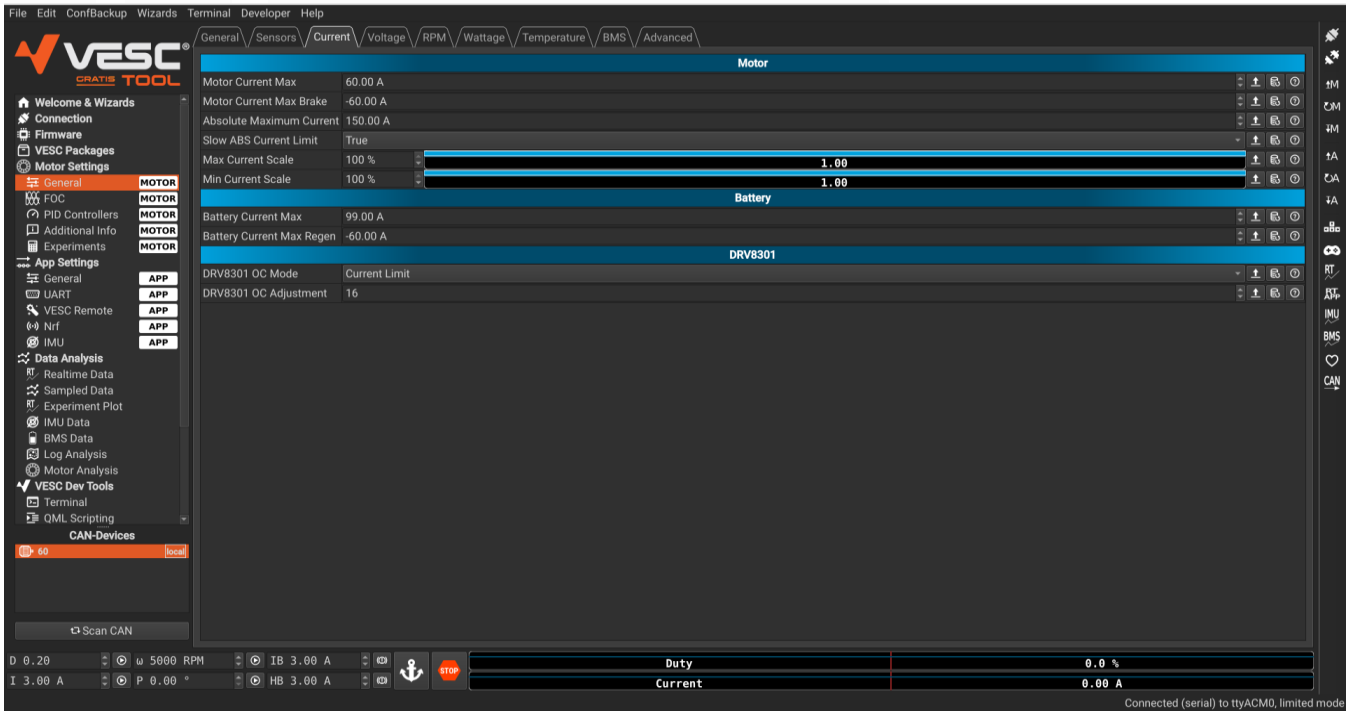


Apply it and press “Write motor configuration”.

Navigate into “Sensorless” tab on the top.
Change the **Openloop Hysteresis** and **Openloop Time** to 0.01, and click the **Write Motor Configuration** button.

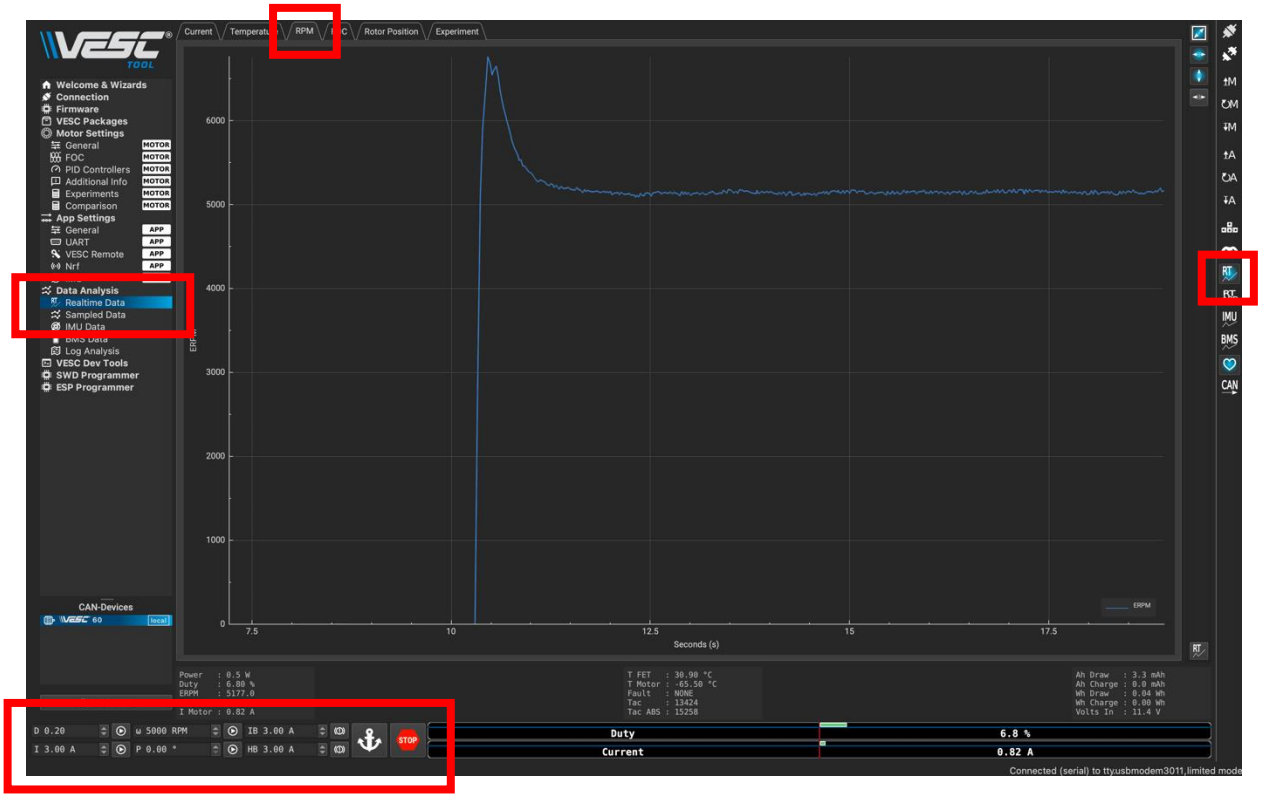


Caution: Additional modifications to other settings are permitted. However, change ‘Motor setting’ ⍰ ‘General’ ⍰ ‘Current’ ⍰ ‘Motor Current Max ' is not allowed for safety.



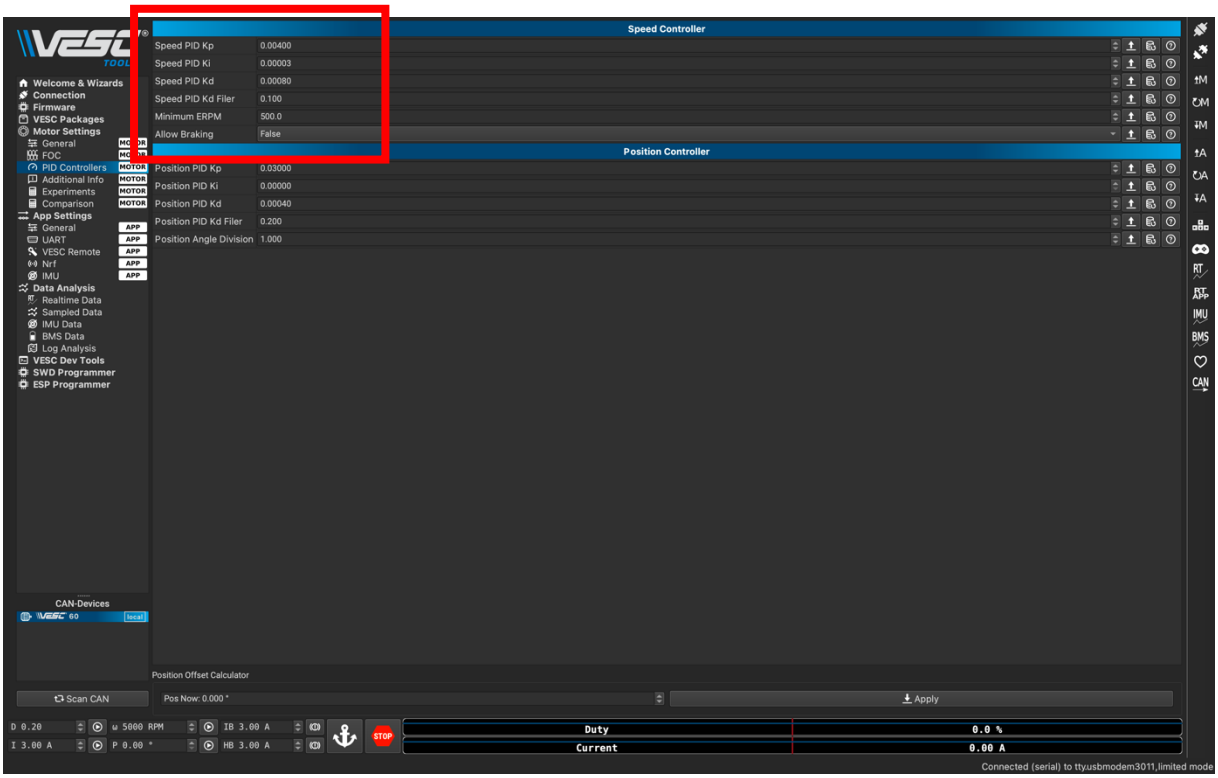
4. PID gain tuning

Now you can start tuning the speed PID controller. To see the RPM response from the motor, navigate to the **Realtime Data** tab under **Data Analysis** on the left. Click **Stream Realtime Data** button on the right (the button with letters RT), and navigate to the **RPM** tab on the top of the screen. You should see RPM data streaming now.



To create a step response for the motor, you can set a target RPM at the bottom of the screen (values between 2000 - 10000 RPM). Click the play button next to the text box to start the motor. Note that the motor will spin, so make sure the wheels of your vehicle are clear from objects. Click the Anchor or STOP button to stop the motor.

You want to look for a clean step response that has a quick rise time and zero to very little steady state error. Adjust the gains accordingly by navigating to the **PID Controllers** tab under **Motor Settings** on the left, and change the Speed Controller gains. General rules of tuning PID gains apply. If you're seeing a lot of oscillations, try changing the Speed PID Kd Filter.



By default, the motor configuration sets a safe top motor RPM. If you wish to change the hard limit set by the VESC firmware, you can go to **Motor Settings** > **General**, and change the max ERPM for forward and backwards rotations. You'll also have to change the configuration file mentioned in the Odometry Tuning section in the software stack setup to change the software limit for your motor ERPM.

